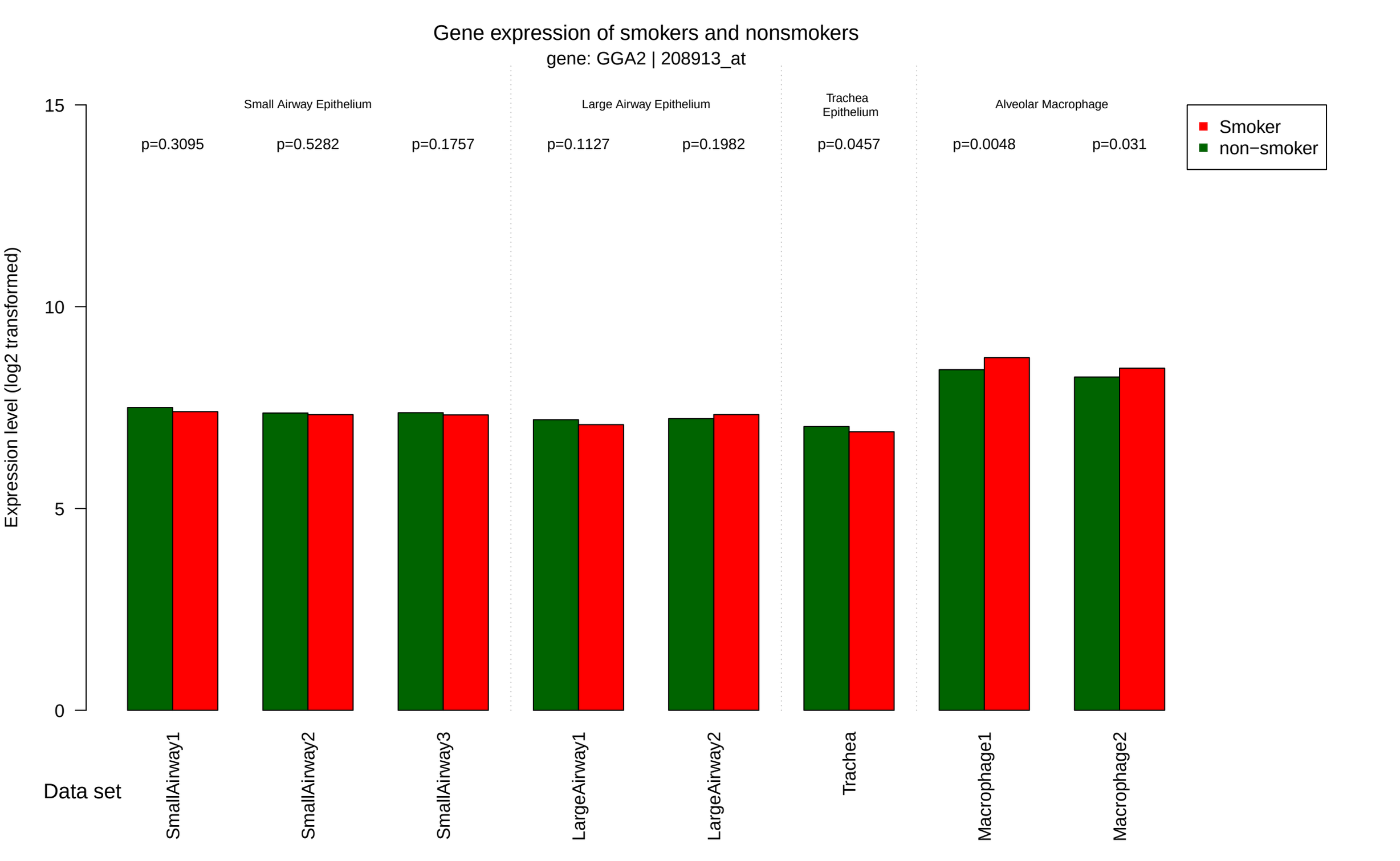
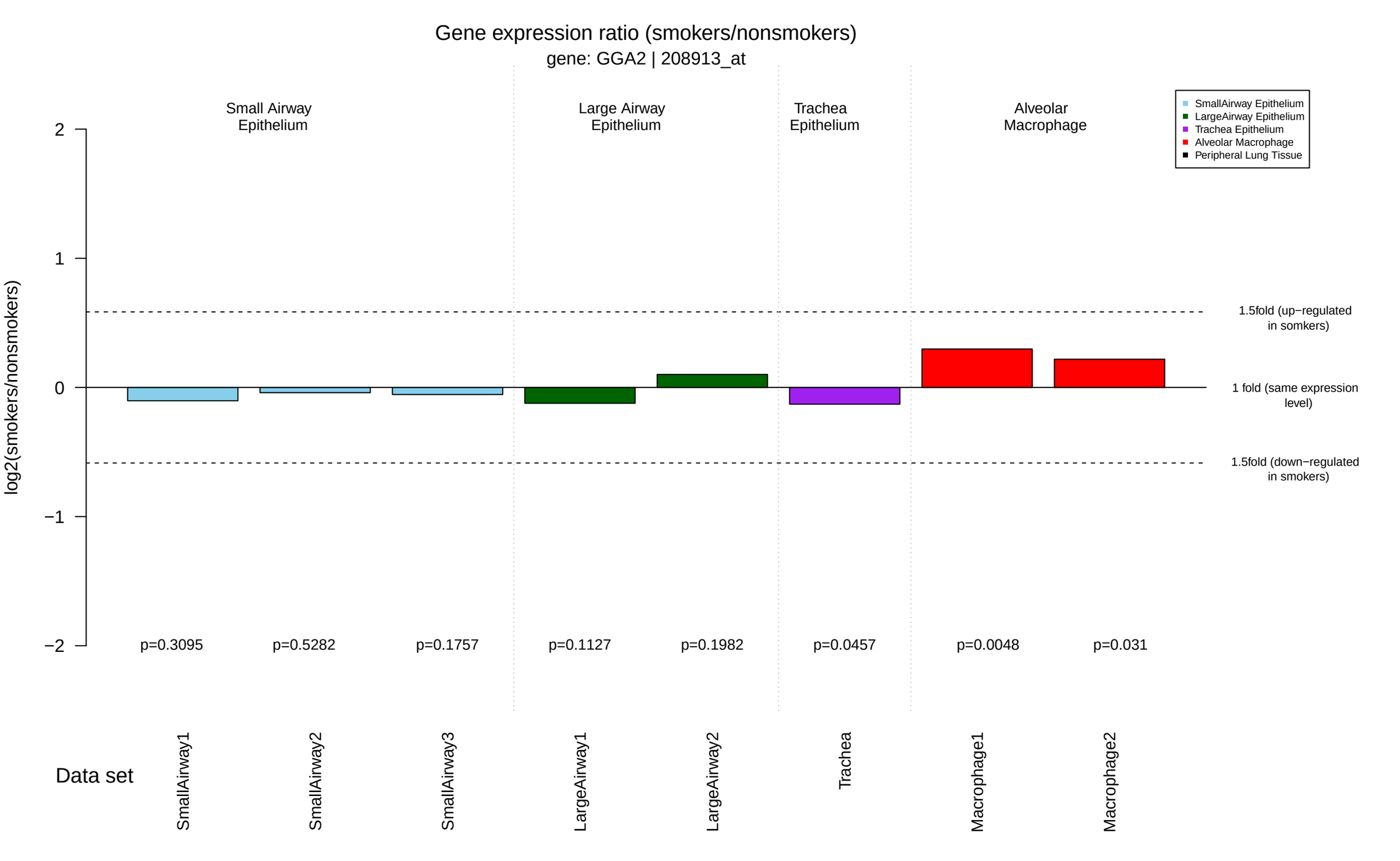
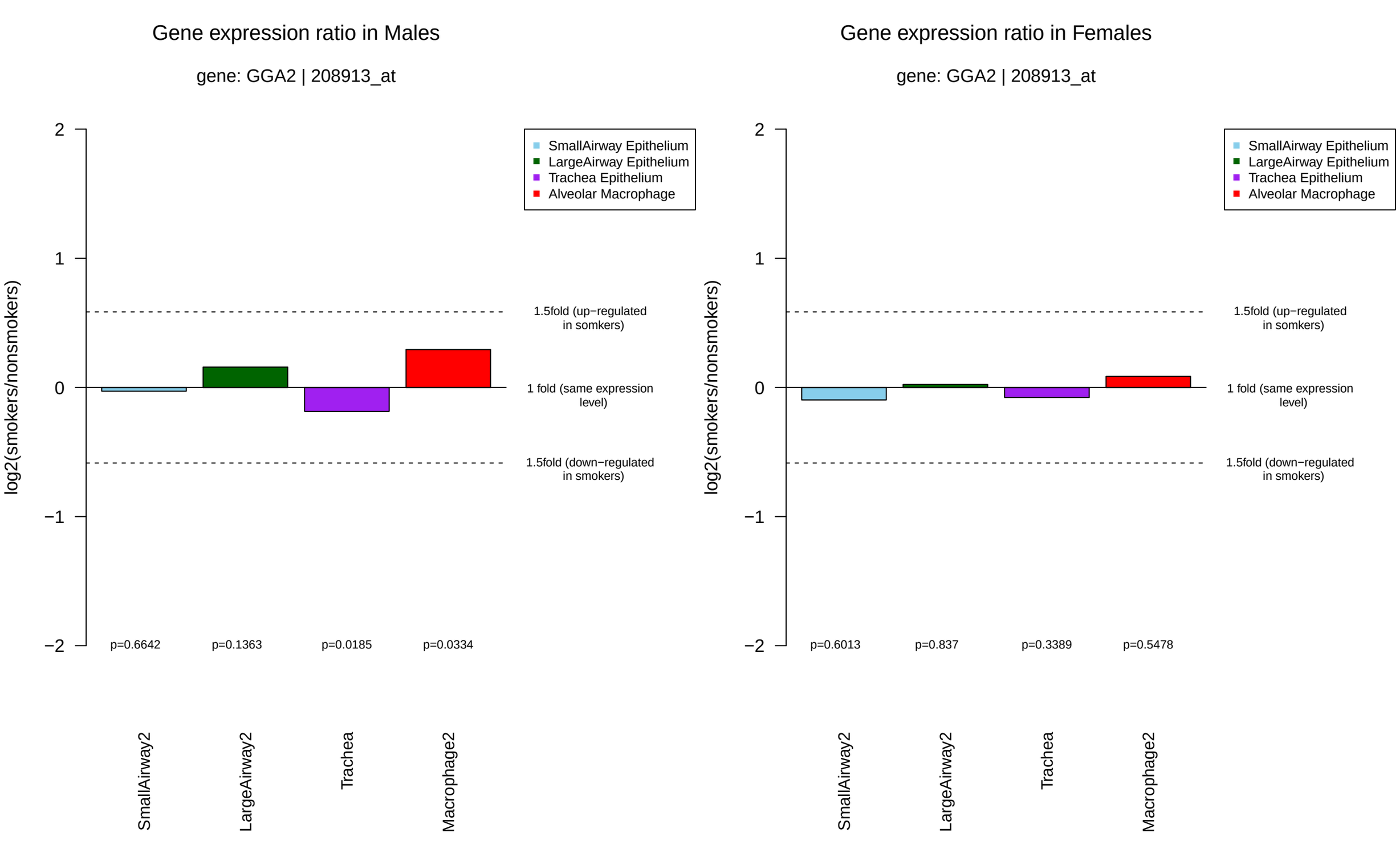
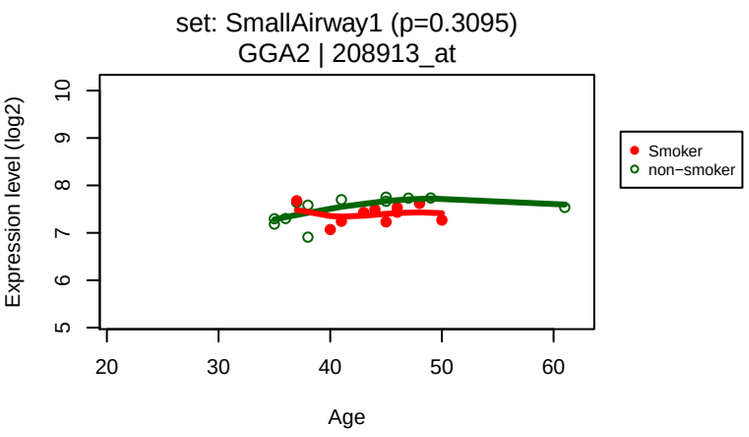
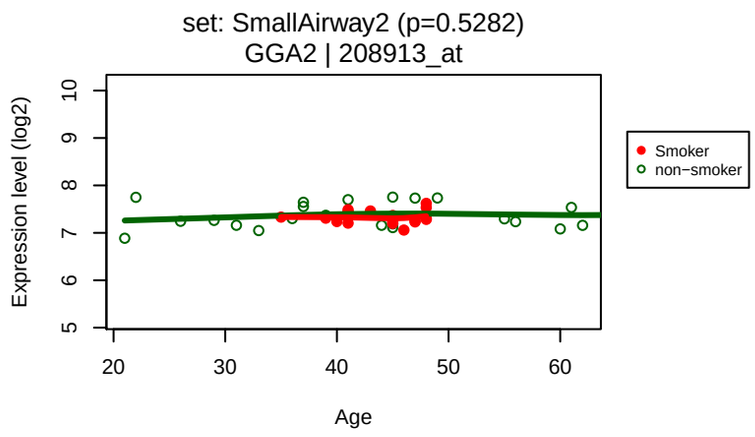
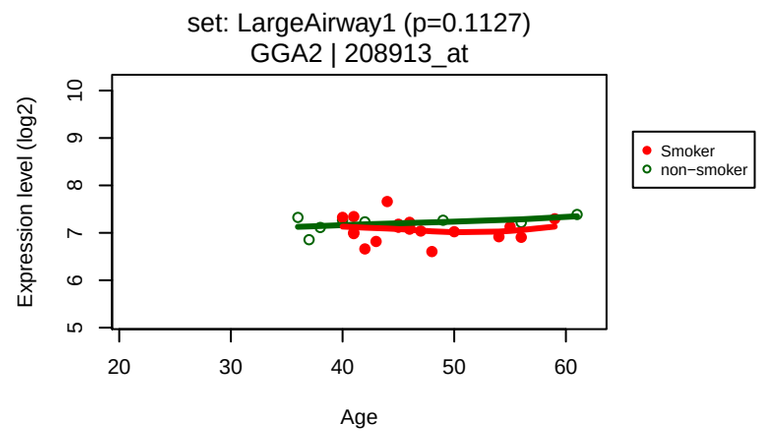

Gene expression of smokers and nonsmokers
gene: GGA2 | 208913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3095
p=0.5282
p=0.1757
p=0.1127
p=0.1982
p=0.0457
p=0.0048
p=0.031
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

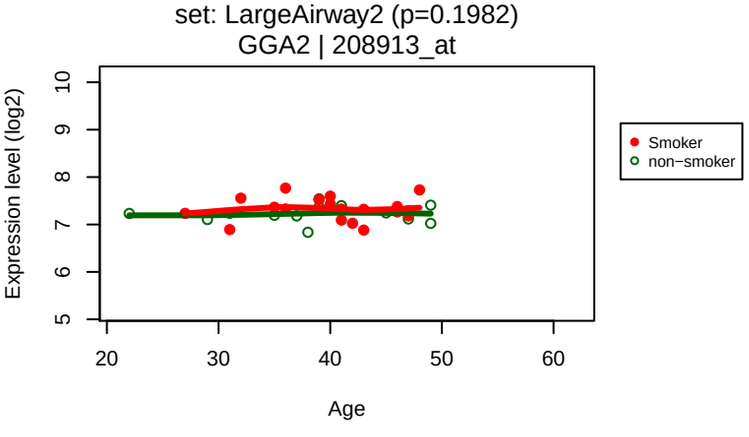

Gene expression ratio (smokers/nonsmokers)
gene: GGA2 | 208913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3095
p=0.5282
p=0.1757
p=0.1127
p=0.1982
p=0.0457
p=0.0048
p=0.031
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

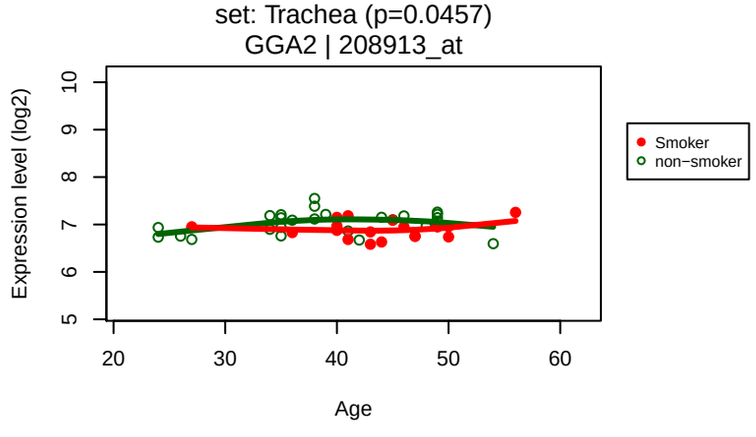

Gene expression ratio in Males
gene: GGA2 | 208913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6642
p=0.1363
p=0.0185
p=0.0334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA2 | 208913_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6013
p=0.837
p=0.3389
p=0.5478
SmallAirway2
LargeAirway2
Trachea
Macrophage2

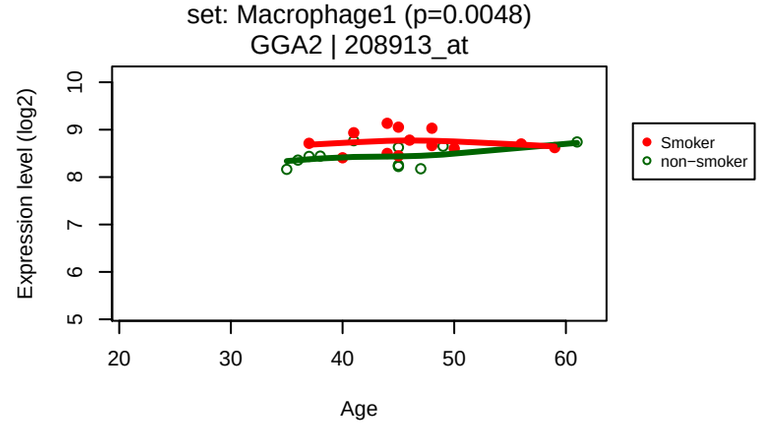

set: SmallAirway1 (p=0.3095)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker

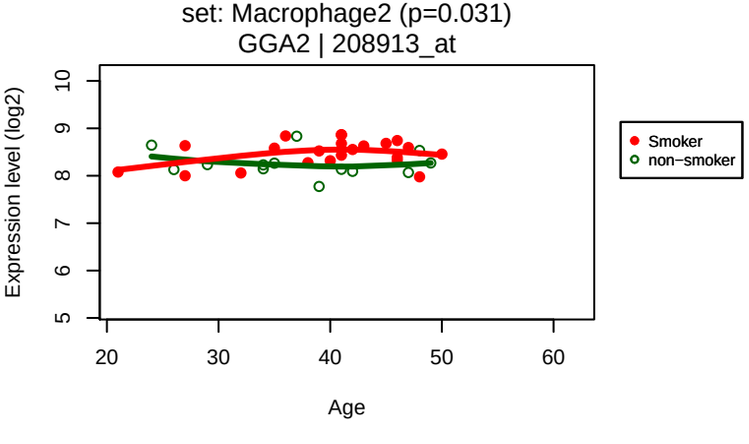

set: SmallAirway2 (p=0.5282)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker

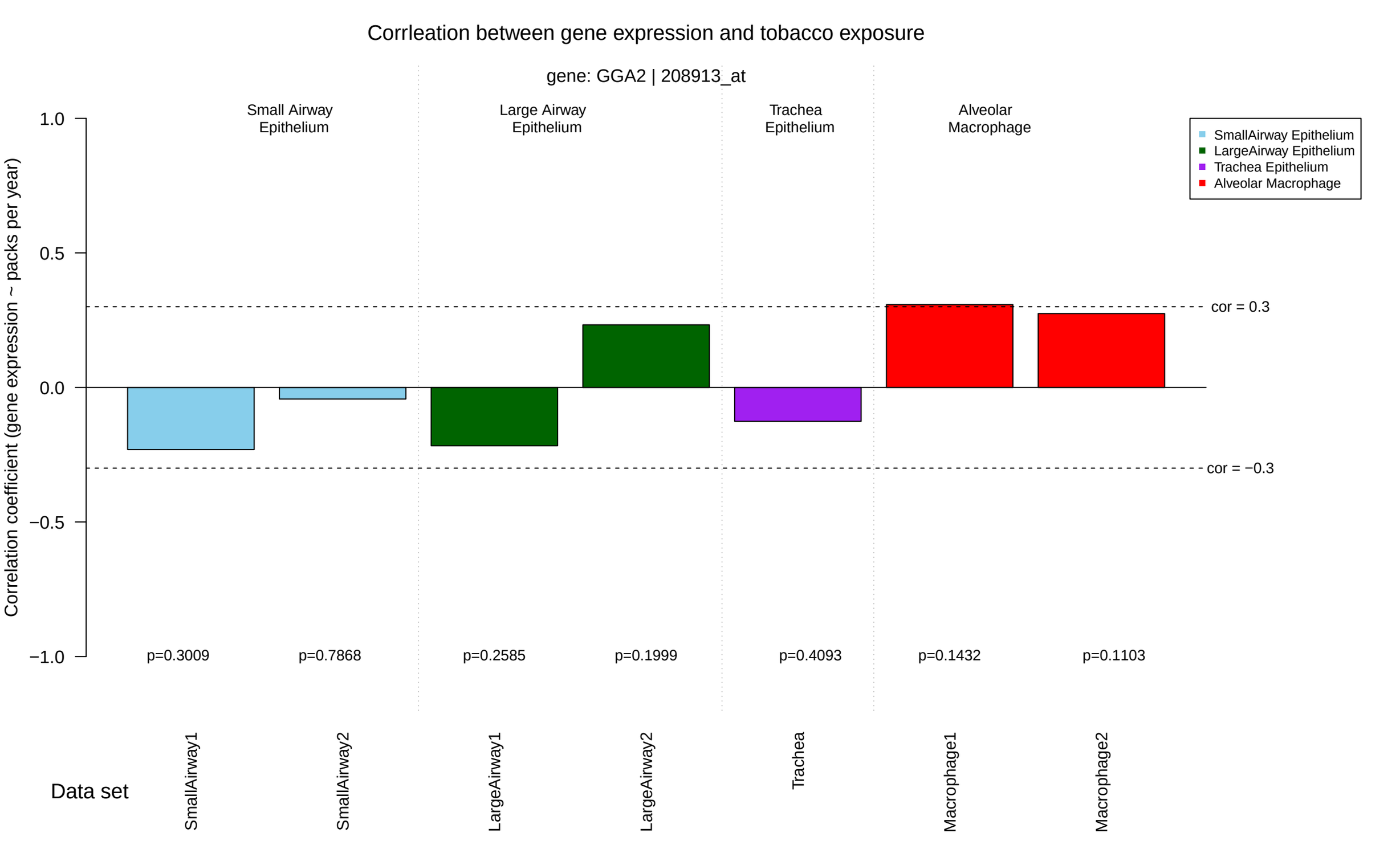

set: LargeAirway1 (p=0.1127)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1982)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0457)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0048)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.031)
GGA2 | 208913_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA2 | 208913_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3009
p=0.7868
p=0.2585
p=0.1999
p=0.4093
p=0.1432
p=0.1103
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2